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Roll No. .....

224415(28)

Diploma in Engg. (Fourth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Elect. Branch)

DIGITAL ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note *: All questions are compulsory, unless mentioned otherwise.*

1. What are the advantages of ditgital system over analog system? What is the chief limitation to the use of digital techniques? 8

2. Perform following : 8

(i) Convert $(5497)_{10}$ to $(\ ?\ )_2$

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(ii) $(47)_{10}$ into gray code.

(iii) $(10011011 \cdot 10111011)_2$ into $(\ ?\ )_8$

(iv) $(3A9E \cdot BOD)_{16} = (\ ?\ )_2$

3. Constract basic gates using universal logic gates. 6

Or

Express the following function as sum of minterms and product of maxterm :

(a) $F(A, B, C, D) = \bar{B}D + \bar{A}D + BD$

(b) $F(x, y, z) = (xy + z)(xz + y)$

4. Reduce the expression $F = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$ and implement the real minimal expression in universal logic gates. 6

Or

Using Boolean Algebra, show that

(i) $D(\bar{A} + B) + B(C + AD) = D + \bar{B}C$

(b) $BC + A\bar{C} + AB + BCD = BC + A\bar{C}$

5. Draw the logic diagram for master-slave J-K flip flop using NAND gate only, write down the truth table and explain it. 6

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6. Compare and contrast sequential circuit and combinational circuit. 6

Or

Write short note on application of flip-flop.

7. (i) Multiplexer is called "Universal Logic Circuit". Justify. 10

(ii) Implement the following Boolean function using 8 : 1 multiplexer.

$$F(P, Q, R, S) = \sum m(0, 1, 3, 4, 8, 9, 15)$$

Or

What is Multiplexer? Draw the logic diagram of 4 : 1 multiplexer using gates and explain its working and also implement 16 : 1 multiplexer using 4 : 1 multiplexer.

8. With the help of logic diagram and truth table explain : 6

(a) 1-line to 4- line demultiplexer

(b) 1-line to 8- line demultiplexer

9. What is difference between decoder and demultiplexer. 4

10. Draw logic diagram for BCD to binary converter using IC 74184 and explain its operation. 10

11. What are output voltage caused by logic 1 in each bit position in an 8-bit ladder if the input level for 0 is 0 V and that for 1 is +10 V. 5

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Or

Define following parameter of DAC's :

(i) Monotonicity

(ii) Offset voltage

(iii) Percentage resolution

(iv) Settling time

(v) Step size

12. Explain successive approximation ADC with its advantage and disadvantage. 5

13. Explain with the help of neat diagram working of MOS RAM cell. 5

Or

Explain difference between SRAM & DRAM.

14. Draw and expalin logic diagram of RAM and EPROM. 5

15. Draw and explain BCD adder using 7483IC. 10

Or

Write short ntoes on : (any **two**)

(i) 1's and 2's complement adder

(ii) Full substractor

(iii) Shift operation

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